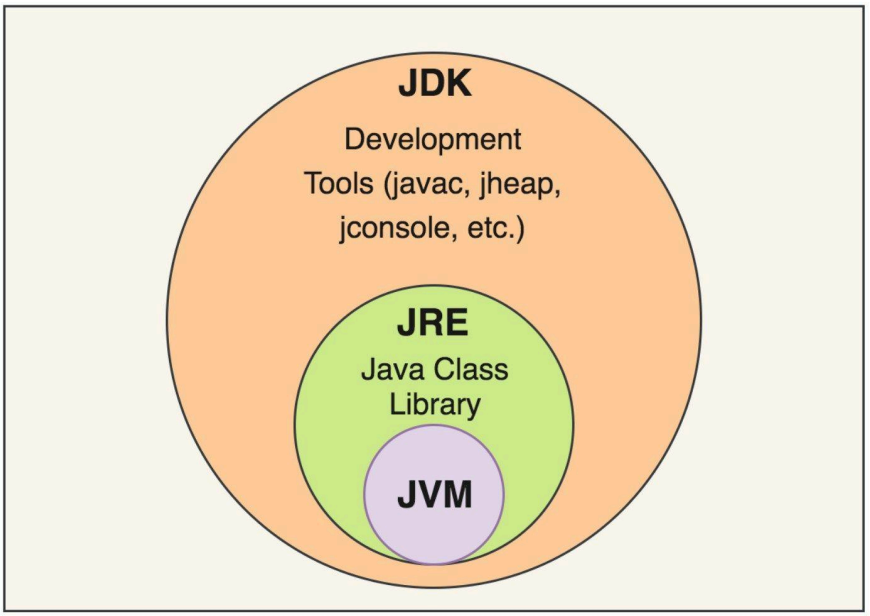
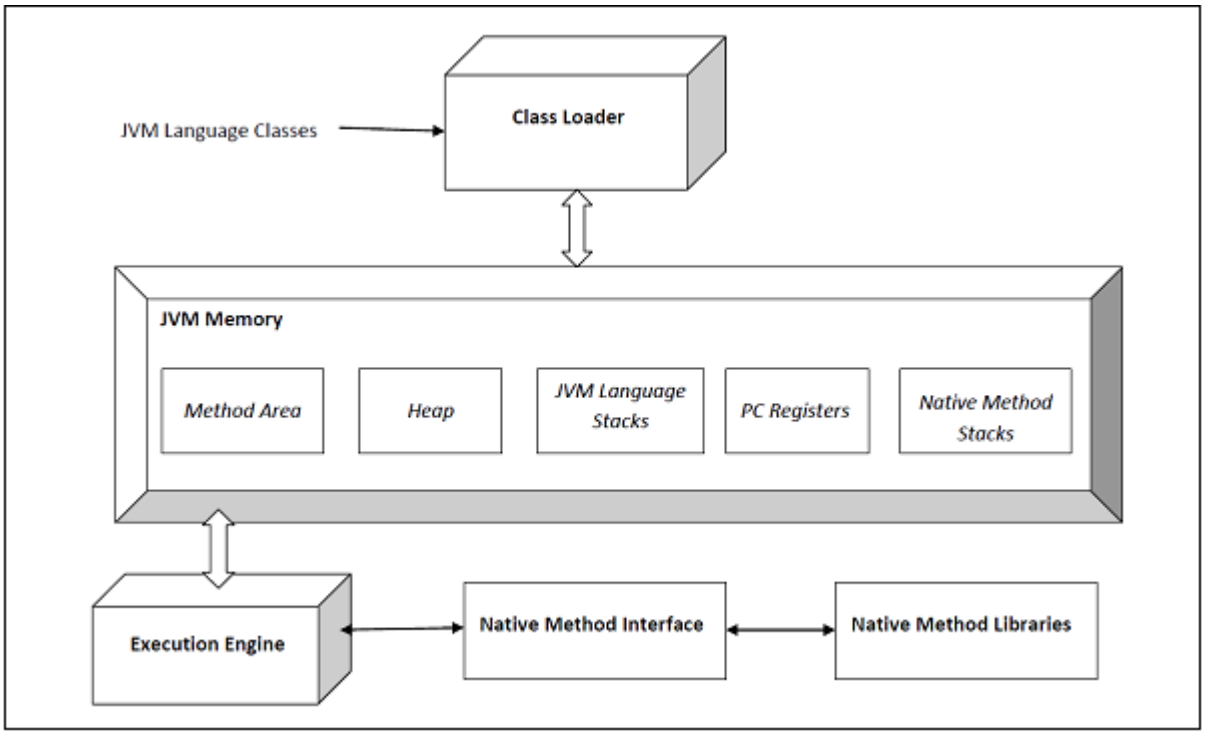
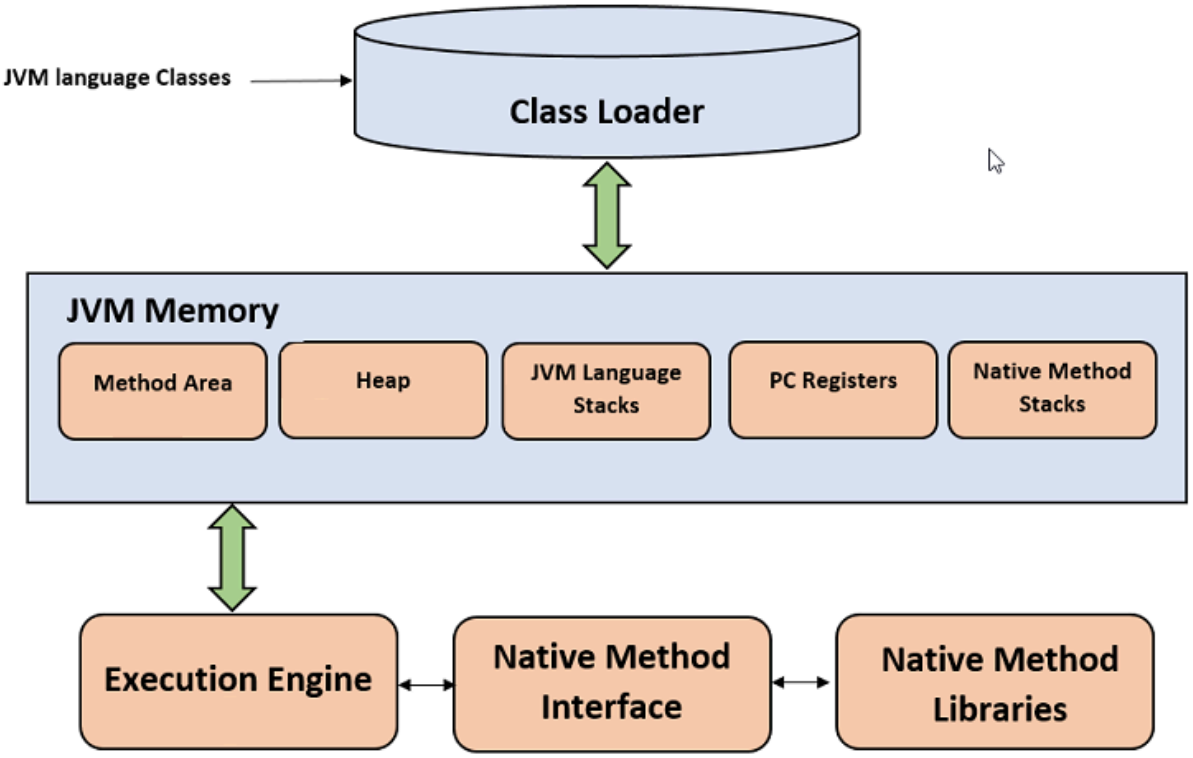
☕ JDK vs JRE vs JVM (Most Asked Java Interview Topic)







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This is one of the **top interview questions** in Core Java.
Let's understand it in the simplest way 👇

🧠 JVM (Java Virtual Machine)

🔹 What is JVM?

JVM is a virtual machine that **runs Java bytecode**.

🔹 Main work:

- Executes Java programs
- Converts bytecode → machine code
- Manages memory
- Garbage collection

👉 JVM makes Java **platform independent**

⚙️ JRE (Java Runtime Environment)

🔹 What is JRE?

JRE provides the **environment to run Java programs**.

🔹 Contains:

- JVM
- Core libraries
- Runtime files

👉 Without JRE, Java programs cannot run.

🛠️ JDK (Java Development Kit)

🔹 What is JDK?

JDK is used to **develop Java applications**.

🔹 Contains:

- JRE
- Compiler (javac)
- Debug tools
- Development utilities

👉 JDK = JRE + Development tools

📊 Relationship (Very Important)

JDK = JRE + Development Tools
JRE = JVM + Libraries
JVM = Runs bytecode

🆚 Difference Table (Interview Ready)

Feature	JVM	JRE	JDK
Full Form	Java Virtual Machine	Java Runtime Environment	Java Development Kit
Purpose	Run bytecode	Run Java program	Develop Java program
Contains	Execution engine	JVM + Libraries	JRE + Compiler + Tools
Required by	End user	End user	Developer

🎤 Interview One-Line Answer

JVM executes Java bytecode, JRE provides runtime environment, and JDK provides tools to develop Java applications.

💡 Easy Real-Life Analogy

- **JDK** → Kitchen (tools to cook)
- **JRE** → Dining table (ready to eat)
- **JVM** → Person who eats (executes program)